■# PAPER<i>336 — g</i>Compressed Complete All-Forces Equation and R(t) 26-Component 4-Subterm Resonant Decomposition

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Session: *95*

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 95 **Source:** gok*share*31b5c807a4.txt (Deep Re-Analysis, Nine-System September 2025 Documents) **Classification:** FIRST g*Compressed complete all-forces form with (M*vis+M_DM) perturbation and fluid buoyancy; FIRST R(t) 4-subterm per state explicit decomposition **Author:** Daniel T. Murphy

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```
F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa =
5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57
```

```
g_\text{UQFF}(r) = g_\text{MUGE}(r)\cdot\Bigl(1 - [SSq]\cdot U_{b_i}\,/\,F_U(r,t)\Bigr),
\quad [SSq] = 0.57
```

Abstract

This paper presents two companion equations from the nine-system September 2025 document assimilation: (1) g*Compressed in its complete all-forces form including the (M*vis+M*DM)(d?/? + 3GM/r^3) dark matter perturbation term, the ?*fluid·V·g fluid buoyancy term, and the quantum Hamiltonian term; and (2) R(t) in its explicit 4-subterm per state decomposition showing all four resonance components: R*U*g1, R*U*g2, R*U*g3, and R*U*g4i per each of the 26 states.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?^{1}, [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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2. g_Compressed Complete All-Forces Equation

2.1 Master Equation

```
g_Compressed(r,t) = (G M(t) / r²) · (1 + H(t,z)) · (1 - B(t)/B_crit) · (1 + F_env(t))
+ ? U_g,i'
+ ? c² / 3
+ ? / v(?x·?p) · ? ?_total† H_op ?_total dV · (2p / t_Hubble)
+ ?_fluid · V · g
+ (M_vis + M_DM) · (d?/? + 3G M / r³)
```

2.2 Term-by-Term Reference

Term 1: Gravitational Base with 3 Multipliers

```
(G M(t) / r²) · (1 + H(t,z)) · (1 - B(t)/B_crit) · (1 + F_env(t))
```

$G M(t)/r^2$: time-evolving Newtonian gravity (M(t) for accreting/star-forming systems)

(1+H(t,z)): Hubble expansion correction at redshift z

(1-B/B*crit): Meissner-type magnetic suppression [from B=0: full gravity; B=B*crit: zero gravity]

(1+F_env(t)): envelope feedback correction

Term 2: Compressed Ug_i Prime Sum

```
? U_g,i' = compressed form of MUGE Ug1+Ug2+Ug3+Ug4 at reduced fidelity
```

Same 26-state structure but with prime notation indicating compression (parameter reduction)

Term 3: Cosmological Constant

```
? c² / 3 = (1.1×10?5² m?²) × (3×108 m/s)² / 3 = 3.30×10?³6 m/s²
```

(Same as PAPER_296 for reference)

Term 4: Quantum Hamiltonian Term (NEW in completeness)

```
? / v(?x·?p) · ? ?_total† H_op ?_total dV · (2p / t_Hubble)
```

?/v(?x?p): Heisenberg uncertainty principle denominator

??†H? dV: quantum expectation value of Hamiltonian

2p/t*Hubble: cosmic-age normalization (PAPER*288 standing-wave bridge)

Term 5: Fluid Buoyancy (NEW in completeness)

```
?_fluid · V · g
```

?_fluid = system-dependent (10?^{2}° to 10?^{1}5 kg/m^3)

V: characteristic system volume

g: local gravity acceleration (self-consistent ? iterative)

Classical Archimedes buoyancy at vacuum fluid density

Term 6: Dark Matter Perturbation Coupling (NEW — not in any prior g_Compressed form)

```
(M_vis + M_DM) · (d?/? + 3G M / r³)
```

Symbol	Value	Description
M_vis	M × f_vis	Visible mass fraction (f_vis=0.15 spiral; 0.05 cluster)
M_DM	M × f_DM	Dark matter mass fraction (f_DM=0.85 spiral; 0.95 cluster)
d?/?	~10?5	Density perturbation parameter
$3GM/r^3$	tidal gravity	3× tidal component

Physical significance: The (M*vis* + *M*DM)(d?/? + $3GM/r^3$) term couples the total mass (visible + dark) to both the density fluctuation field AND the tidal gravity — simultaneously handling dark matter dynamics AND structure formation in one term. This is the UQFF generalization of the Jeans instability criterion and the dark matter halo perturbation.

2.3 Results by Scale Class

System	g_Compressed (N)	Dominant Terms
Vela (compact)	~3.95×10?4^{1}	Term 1 × Hubble + Term 4

System	g_Compressed (N)	Dominant Terms
Crab (compact)	~3.95×10?4[1]	Term 1 + Term 3
NGC 1365 (galactic)	~4.12×10?4[1]	Term 6 (DM) + Term 1
Abell 2256 (cluster)	~4.12×10?4[1]	Term 6 (DM dominant)
Jupiter	~3.95×10?4[1]	Term 1 (giant planet regime)

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3. R(t) 26-Component 4-Subterm Resonant Decomposition

3.1 Master Equation

```
R(t) = ?_{i=1}^{26} [ R_U_g1,i · cos(?_U_g1,i · t)
+ R_U_g2,i · cos(?_U_g2,i · t)
+ R_U_g3,i · cos(?_U_g3,i · t)
+ R_U_g4i,i · cos(?_U_g4i,i · t) ]
```

3.2 Four Resonance Sub-Terms per State

Each of the 26 states i contributes 4 cosine resonance components:

Sub-term	Physical Origin	Frequency Scale
R*U*g1,i · cos(?*U*g1,i t)	Magnetic dipole resonance	f_super = 1.411×10[1]6 Hz at i=1
R*U*g2,i · cos(?*U*g2,i t)	Charge-reactivity resonance	f_react = 10[1°] Hz at i=1
R*U*g3,i · cos(?*U*g3,i t)	String rotation resonance	f_THz = 10^{12} Hz at i=1
R*U*g4i,i · cos(?*U*g4i,i t)	Vacuum concentration resonance	f_quantum = 1.445×10?[1]7 Hz at i=1

State-to-state scaling: ?*U*gX,i decreases with increasing i by the [SSq] exponential factor:

```
?_U_gX,i = ?_U_gX,1 × exp(-[SSq] · i/26)
```

3.3 Total Term Count

- 26 states × 4 sub-terms = 104 individual cosine terms
- Each term carries amplitude R*U*gX,i ~ f*X* × *A*X(state)
- TRIADIC master (PAPER_326) showed the 26-state structure; THIS paper shows the 4-subterm internal structure

3.4 Results by Scale Class

System	R(t) (N)	Dominant Sub-term
Vela/Crab (compact)	-1.12×10?4[2]	R*U*g3 THz (f_THz=10^{12} blob velocities 0.3-0.7c)
NGC 1365/ESO 137-001	-2.29×10?4[1]	R*U*g1 magnetic dipole (Seyfert AGN)
Jupiter/Lagoon	-1.12×10?4[2]	R*U*g2 charge-reactivity (H3+/ionized plasma)

3.5 Vela Frequency Assignment

For Vela Pulsar THz blobs (0.3–0.7c velocities, f_res~10^{12} Hz):

```
R_U_g3,i = R_0 · (v_blob/c) · (f_THz/f_ref) [THz component dominant]
```

```
?_U_g3,i = 2p × f_THz = 2p × 10¹² rad/s
```

The ~0.3 phase separation characteristic of the Vela multi-peak profile maps to:

```
phase_sep = 0.3 ? cos(p × 0.3/0.3) = cos(p) = -1 [anti-phase sum]
R_total ~ 2 × |R_U_g3| × cos(p × phase_sep) at minimum
```

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4. Integration Context

g_Compressed and R(t) are two of the four UQFF modes:

g_Compressed — this paper (Term 6 DM perturbation NEW)

R(t) — this paper (4-subterm per state NEW)

F*U*Bi — PAPER_335 (buoyancy kernel)

U_i — PAPER_334 (superconductive vacuum density)

Together they form the complete MUGE-to-FLENR decomposition:

```
g_total = g_Compressed + R(t) + F_U_Bi/m + F_U_Bi_i/m
```

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5. FIRST Declarations

FIRST g_Compressed complete all-forces form — includes Term 6: (M_vis+M_DM)(d?/? + 3GM/r³) dark matter perturbation

FIRST fluid buoyancy term (?*fluid·V·g) in g*Compressed reference

FIRST R(t) 4-subterm per state explicit decomposition — R_Ug1/Ug2/Ug3/Ug4i per each of 26 states (104 total terms)

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6. Key Equations Summary

```
g_Compressed = (GM(t)/r²)(1+H(t,z))(1-B/B_crit)(1+F_env)
+ ?U_g,i' + ?c²/3
+ ?/v(?x?p)·??†H_op ? dV·(2p/t_Hubble)
+ ?_fluid·V·g
+ (M_vis+M_DM)·(d?/? + 3GM/r³) [NEW DM perturbation]
R(t) = ?_{i=1}^{26} [R_Ug1,i cos(?_Ug1,i t) [magnetic dipole]
+ R_Ug2,i cos(?_Ug2,i t) [charge-reactivity]
+ R_Ug3,i cos(?_Ug3,i t) [string rotation ? THz]
+ R_Ug4i,i cos(?_Ug4i,i t)] [vacuum concentration]
[compact] g_Compressed ˜ 3.95×10?4¹ N; R(t) ˜ -1.12×10?4² N
[galactic] g_Compressed ˜ 4.12×10?4¹ N; R(t) ˜ -2.29×10?4¹ N
```

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7. References

gok*share*31b5c807a4.txt (September 14, 2025 — 9-document assimilation)

Vela Pulsar (PSR J0835-4510 in Vela Remnant)_12Sept2025.docx — Compressed + Resonant eqs 1-5

NGC 1365 (Great Barred Spiral Galaxy)_12Sept2025.docx — eqs 6-10

Abell 2256 (Galaxy Cluster)_11Sept2025.docx — eqs 16-20

PAPER_326: Triadic Master (R(t) 26-state structure; structural predecessor)

PAPER_288: Cosmic-Age Standing-Traveling Wave Bridge (quantum Hamiltonian term context)